Experiment 5:

Code:

```
  GNU nano 9.0
#include <stdio.h>

int main()
{
char str[100];
int start, length, i, count = 0;

printf("Enter a string: ");
fgets(str, 100, stdin);
while (str[count] != '\0' && str[count] != '\n')
{
count++;
}
printf("Length of the string: %d\n", count);

printf("Enter starting position: ");
scanf("%d", &start);
printf("Enter number of characters: ");
scanf("%d", &length);

printf("Substring: ");

for(i = start; i < start+length && i < count; i++)
{
printf("%c", str[i]);
}
printf("\n");
return 0;
}
```

Output:

```
┌──(kali㉿sourav)-[~]
└─$ time ./exp5
Enter a string: Laplus
Length of the string: 6
Enter starting position: 0
Enter number of characters: 5
Substring: Laplu

real    7.93s
user    0.00s
sys     0.00s
cpu     0%
```

Experiment 5:

Code:

```
  GNU nano 9.0
#!/bin/bash

read -p "Enter a string: " str

echo "Length of the string: ${#str}"

read -p "Enter starting position:" start
read -p "Enter number of characters: " length

echo "Substring: ${str:$start:$length}"
```

Output:

```
┌──(kali㉿sourav)-[~]
└─$ time bash exp5a.zsh
Enter a string: Laplus
Length of the string: 6
Enter starting position:0
Enter number of characters: 6
Substring: Laplus

real    11.96s
user    0.00s
sys     0.00s
cpu     0%
```